

PAPER 02 ZERO PARAMETER ROTATION CURVE FITTING SPARC 175 ANALYSIS

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SUBSTRATE NOTICE (2026-09-20). This document states its kinematic substrate on the big-dimension frame $V_{240}(\mathbb{R}^{57,600})$, which was **retired 2026-08-25**. The live substrate is $V_2(\mathbb{R}^3)$ via Cartan triality. The retirement was structural rather than cosmetic, so this text needs a rebuild and not a patch; it is retained unaltered as a record of the earlier formulation. The arithmetic $240^2 = 57,600$ remains a true E_8 root-count identity, but it is not the ambient dimension of anything in the current theory. No migrated $V_2(\mathbb{R}^3)$ derivation is claimed here. **Superseded framing (2026-10-09).** The title’s “zero parameter” claim is withdrawn. The model carries two irreducible inputs, the a_0 scale and the choice of μ ; current SPARC numbers are generated in `SPARC_PARAMETER_BUDGET.md`.

Abstract

Manuscript part of the 42-Paper Canonical Corpus of Geometrically Ordered Dynamics.

1 Introduction

SUPERSEDED (2026-08-25). The substrate identification $V_{240}(\mathbb{R}^N)$ below is stated on the retired big-dimension frame $V_{240}(\mathbb{R}^{57,600})$. The live substrate is $V_2(\mathbb{R}^3)$ via Cartan triality; the big-dimension frame is retired. The arithmetic $240^2 = 57,600$ remains true as an E_8 root-count identity, but its identification as the ambient substrate dimension is not current canon. No migrated $V_2(\mathbb{R}^3)$ derivation is claimed; the original text is retained unaltered below as a record.

This manuscript formalizes domain invariants under chiral alignment constraint $\chi \geq 0.7$ on the Stiefel manifold $V_{240}(\mathbb{R}^N)$.

2 Core Mathematical Derivation

$$\mathcal{S}_{eff} = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R + \mathcal{L}_{IT}(\chi) \right] \quad (1)$$

Theorem 1 (Domain Core Invariant). Continuous alignment path $\text{Hol}(\omega) \in SO^+(240)$ holds for all t .

3 Conclusion

This completes the formalization for this domain within the G.O.D. framework.